

to juggle with shell scripts to modify files for readability, then Python for preprocessing, then C for computation-intensive tasks. Pick one learning curve - the Intel® DAAL curve, and follow it to fruition. That's all you'll need.

IT'S REALLY FAST

'Significantly faster' is all very well, but how fast? Hold on to your seats, because Intel® DAAL overdelivers. Here are some metrics of its performance in comparison with long-time favourite Apache Spark's MLlib.

And finally, it is the turn of Pythonistas to be awestruck. They have always known that Python is a language of human-side elegance and readability - sacrificing performance for it's beauty and ease-of-scripting. But the 100x to 1000x performance edge Intel® DAAL adds to scikit-learn is hard to ignore.

THE INTEL® DAAL WORKFLOW

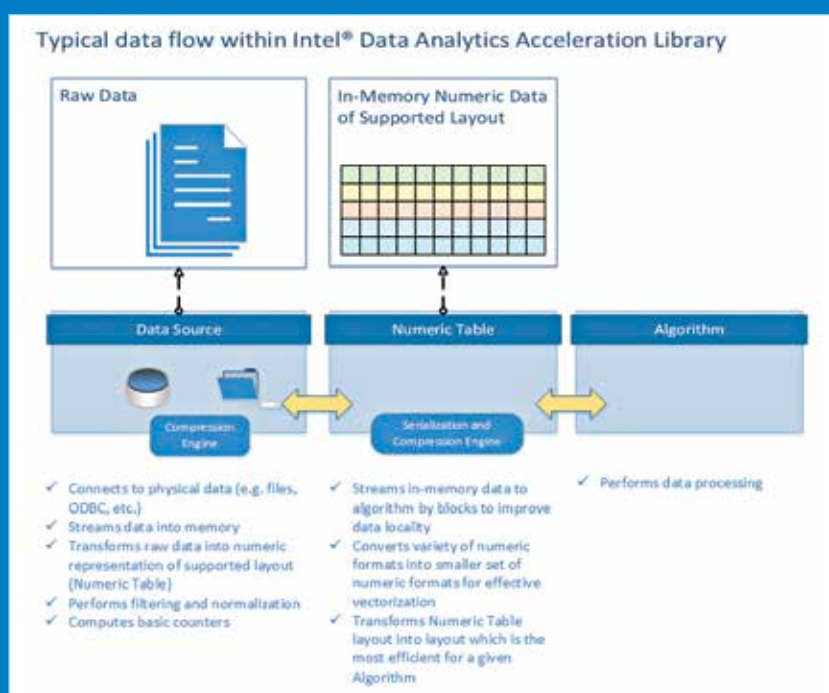
Intel® DAAL has a simple and robust workflow that can connect to a versatile range of existing systems. It has an efficient architecture for reading and storing data from a variety of sources, be they raw files, ODBC, SQL databases, or almost anything else. There's also efficient and useful metadata and basic counter computation - this allows you to get a sense of your data before you even begin to analyse it. The principles of data exploration guiding analysis are well-integrated with Intel® DAAL.

An important component of Intel® DAAL's performance advantage is the efficient vectorization and block-wise storage of data to suit a variety of algorithms. Not only is general storage extremely well-designed but Intel® DAAL can also dynamically preprocess inputs to speed up specific algorithms in certain ways. For example, it adapts Numeric Tables to a layout that optimizes space- and time-efficiency for the required algorithm, just before beginning computation.

Intel® DAAL also serves diverse compute models, like batch, streaming, or distributed systems. This creates flexibility in dealing with data sets that have different storage and performance needs. Intel® DAAL's mathematical subroutines are implemented with a bottom-up approach for maximum efficiency in incremental processing, master-slave distribution of tasks, as well as a well-integrated API to call these methods.

PUTTING YOUR MONEY WHERE YOUR MOUTH IS

Intel® DAAL offers a host of popular algorithms. From computing quantiles, covariance matrices, statistical moments, and decompo-



A schematic of the Intel® DAAL workflow

sitions, to building Naive Bayes classification models, different kinds of regressions, neural networks, and decision forests, Intel has thought of it all. It's no small wonder that Intel is quickly adopting a 'join us or fight us' strategy in these markets. It simply makes sense to integrate Intel's insight with standard libraries, as Python's scikit-learn has already begun doing.

Intel® DAAL also offers developers the freedom to leverage the implementation of various data structures, subroutines and algorithms to build their own algorithms and methods for specific purposes. In this, Intel has shown its commitment to open-source culture, a bold gesture that could have a significant impact on data science in the future.

GETTING YOUR HANDS ON INTEL® DAAL

By now, you're probably convinced. You're ready to give Intel® DAAL a spin, despite the existential dread brought by the prospect of change. A hundred times faster, you're telling yourself. A hundred times more expensive, you might be dreading.

In software, the best things often come at a hefty price, contrary to popular wisdom. However, Intel® DAAL is currently free. For a product in its relative infancy, it might pay to be an early adopter, before Intel starts cornering the market.

Intel® DAAL is completely open source, requires no royalties to use or even to redistribute, and has a community of helpful users that's growing every day. Intel offers a free li-

cense for Intel® DAAL that allows you access to all of these features, as well as open community support. There's also a paid license - at \$499 for the 1st year, and \$179 for timely renewal thereafter, users get access to private support from Intel's engineers. This might be a worthwhile investment for corporate use cases. Adopters of Intel® DAAL for academic research have access to a lower slab of \$249 for the first year, and a discounted renewal rate. Note that renewal of the license is more expensive if you wait for the previous one to expire.

GETTING STARTED

Intel® DAAL (whether stand-alone, or in conjunction with Intel's other offerings) is free to download. Its Python, C++ or Java APIs are easy to install - though optimal performance with your hardware may require a separate installation of the Intel® DAAL libraries. There's a wealth of documentation that isn't just boring black-and-white man pages. Intel has built a useful collection of gentle introductions to topics in data analysis, a number of blog posts in different degrees of technical detail, and lots of helpful community discussions. Intel has a lot more developer resources for AI and other applications so if you are curious, head over to <https://software.intel.com/en-us/>.

Whether a top-notch implementer of ML solutions, a researcher in the field, or simply an interested dabbler, there are a lot of reasons to start tinkering with Intel® DAAL. Go forth, and may the gradients be with you. ■